

AI Internship Recommendation Engine

Detailed Project Report

Task ID: AI-SS-002

Domain: Student Support & Internship Management Recommendation

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Repository: github.com/anshuldeepbajpai-dhoni/AI-Internship-Recommendation-Engine

A hybrid machine-learning recommendation system that combines collaborative filtering and content-based matching to rank internships for existing and cold-start students.

1. Executive Summary

The AI Internship Recommendation Engine is designed to personalize internship discovery for students. Instead of presenting the same internship list to every user, the system uses student profile attributes and historical interaction patterns to rank opportunities according to likely relevance. The implementation contains three recommendation approaches: user-based collaborative filtering, TF-IDF content-based filtering, and a hybrid model that combines both signals.

The project also addresses the cold-start problem. A new student without previous internship ratings can still receive recommendations from skills, interests, preferred domain, and experience level. A Streamlit dashboard exposes the recommendation workflow and analytics, while automated Pytest coverage validates collaborative, content, hybrid, cold-start, evaluation, exclusion, and weighting behavior. The supplied test screenshot shows 12 collected tests and 12 successful passes.

Component	Implementation
Language	Python 3.11
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
Similarity	Cosine similarity
Content Representation	TF-IDF
Recommendation Models	Collaborative, Content-Based, Hybrid
Interface	Streamlit
Testing	Pytest
Data Source	CSV datasets for students, internships, and interactions

2. Problem Statement and Objectives

Students often need to search through many internship opportunities while manually judging whether the required skills, domain, and experience level match their profile. The project formulates this as a recommendation problem: given a student and an internship catalog, rank unseen internships according to predicted relevance.

The main objectives are to preprocess student, internship, and interaction data; create a user-item rating matrix; identify similar students; model profile-to-internship similarity; combine collaborative and content signals; support new students without ratings; generate Top-N recommendations; evaluate recommendation behavior; and provide an interactive dashboard for demonstration and analysis.

3. Data and Preprocessing Pipeline

The application works with three logical datasets. Student profiles provide identifiers and descriptive attributes such as skills, interests, domain, and experience level. Internship profiles contain identifiers, title, company, domain, duration, required skills, and experience information. Interaction records connect students to internships through ratings. The preprocessing layer standardizes text values, parses skill fields into usable lists, validates records, and prepares the interaction data for matrix construction.

The user-item matrix places students on rows and internships on columns, with ratings representing known preferences. Unrated entries provide the candidate space for collaborative recommendation. This representation is central to user-based collaborative filtering because similarity can be computed from overlapping preference patterns.

4. Recommendation Methodology

4.1 Collaborative Filtering

The collaborative model uses student rating behavior rather than profile text. Cosine similarity is computed between students in the user-item matrix. For a target student, similar students act as neighbors, and their ratings for unseen internships contribute to similarity-weighted predicted scores. Previously interacted internships are excluded from the final candidate list.

4.2 Content-Based Filtering

The content model builds textual representations of both sides of the matching problem. Student profiles combine skills, interests, domain, and experience level. Internship profiles combine required skills, domain, experience level, and title. TF-IDF transforms the internship corpus into vectors and the student profile into the same feature space. Cosine similarity then measures profile-to-internship relevance, producing a score between 0 and 1.

4.3 Hybrid Recommendation

The hybrid recommender combines the strengths of the two models. Collaborative rating predictions are normalized to a 0–1 scale so they are comparable with content similarity. Existing students use a weighted score of 60% collaborative signal and 40% content signal. For a cold-start student with no interaction history, the content model becomes the usable signal and provides recommendations directly rather than penalizing the student for missing ratings.

5. Evaluation Strategy

The evaluation utilities implement Precision@K and Recall@K for Top-N ranking quality, together with MAE and RMSE utilities for rating-prediction error. Leave-one-out evaluation can remove a relevant interaction from training, rebuild the model, and test whether the held-out internship appears among the recommendations. Because the supplied screenshots do not show the actual Precision@5, Recall@5, MAE, or RMSE outputs, this report does not invent numerical values for those metrics.

6. Streamlit Dashboard and System Demonstration

The dashboard provides four navigation areas: Home, Existing Student, New Student, and Analytics. The following figures are taken directly from the supplied application screenshots.

6.1 Home Dashboard

The Home view presents the project purpose, dataset-level KPIs, the three recommendation approaches, and the internship catalog. The supplied screenshot shows 10 students, 12 internships, and 28 interactions. It also explains Collaborative, Content-Based, and Hybrid recommendation at a glance.

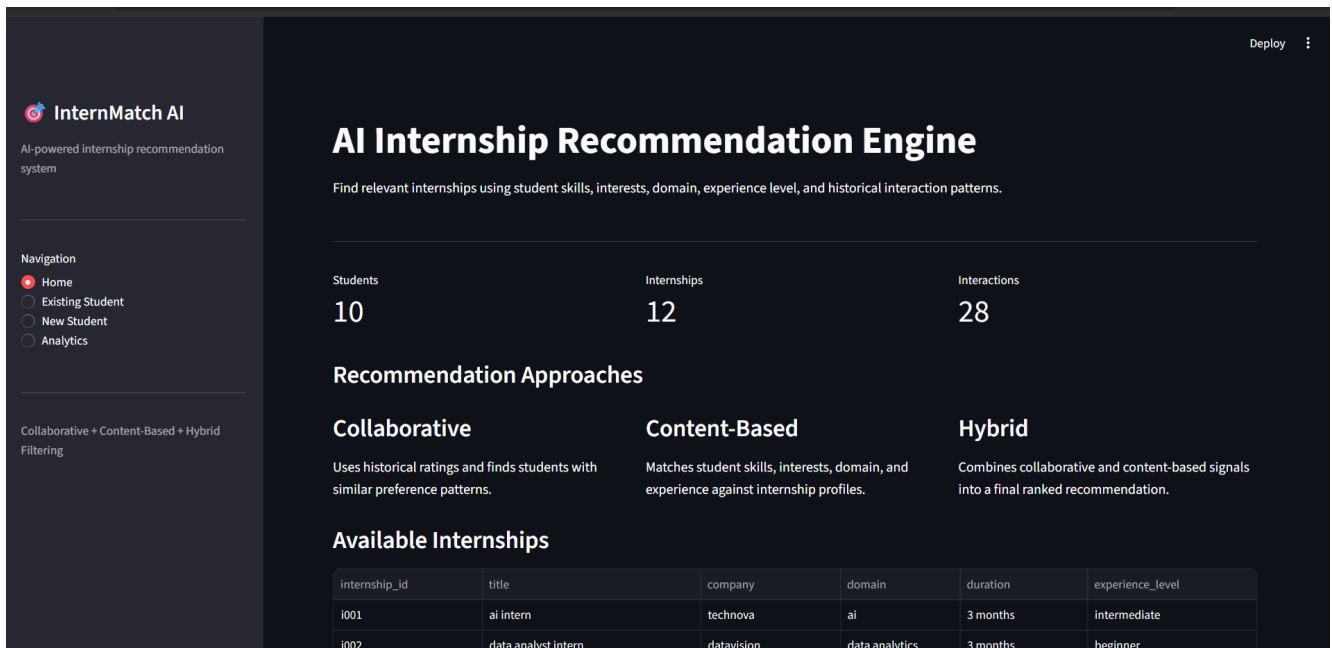


Figure 1: InternMatch AI – Home Dashboard and Recommendation Approaches

6.2 Existing Student Recommendation Workflow

The Existing Student page lets the user select a registered student, inspect profile information, choose a recommendation method, set the number of recommendations, and generate results. In the supplied screenshot, student S001 — Aarav has domain AI, intermediate experience, three interactions, and skills including Python, Pandas, Machine Learning, and Data Science. The selected recommendation method is Hybrid with five requested recommendations.

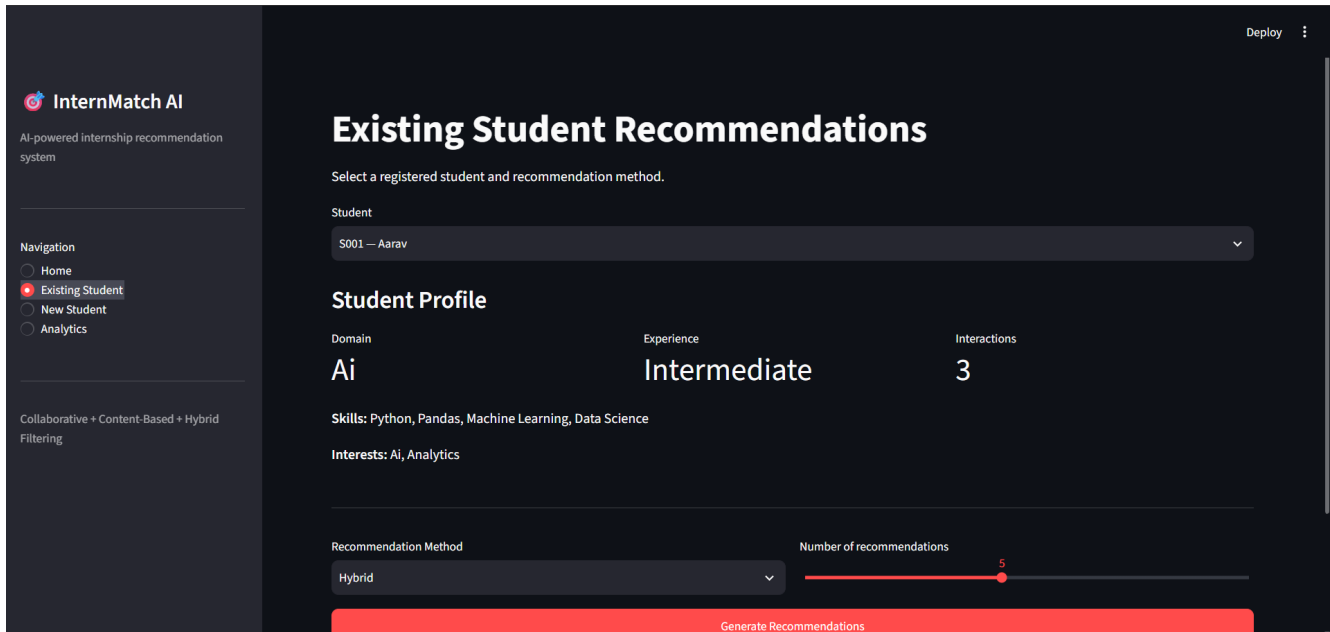


Figure 2: Existing Student Profile and Recommendation Configuration

6.3 New Student and Cold-Start Handling

The New Student page demonstrates cold-start support. The user can provide skills, interests, preferred domain, experience level, and Top-N count without supplying any historical rating. The displayed form is configured for an AI-oriented beginner profile and supports direct generation of internship matches from profile content.

The screenshot shows the 'New Student Recommendation' interface. On the left is a sidebar with the 'InternMatch AI' logo, a description 'AI-powered internship recommendation system', and a navigation menu with 'Home', 'Existing Student', 'New Student' (selected), and 'Analytics'. Below the menu is the text 'Collaborative + Content-Based + Hybrid Filtering'. The main content area has a title 'New Student Recommendation' and a subtitle 'Enter a student profile to receive recommendations without requiring previous internship ratings.' Below this is a form with the following fields: 'Skills' (text input with 'Python, Pandas, SQL, Machine Learning'), 'Interests' (text input with 'AI, Data Science, Analytics'), 'Preferred Domain' (dropdown menu with 'ai' selected), 'Experience Level' (dropdown menu with 'beginner' selected), and 'Number of recommendations' (a slider set to 5). At the bottom of the form is a red button labeled 'Find Internships'. In the top right corner of the main area is a 'Deploy' button and a menu icon.

Figure 3: New Student Cold-Start Recommendation Interface

6.4 Recommendation Analytics

The Analytics view summarizes the dataset and visualizes recommendation-related data. The supplied dashboard reports 10 students, 12 internships, 28 interactions, and an average rating of 4.36. The domain chart shows the internship catalog distributed across AI, data analytics, mobile, and web domains.

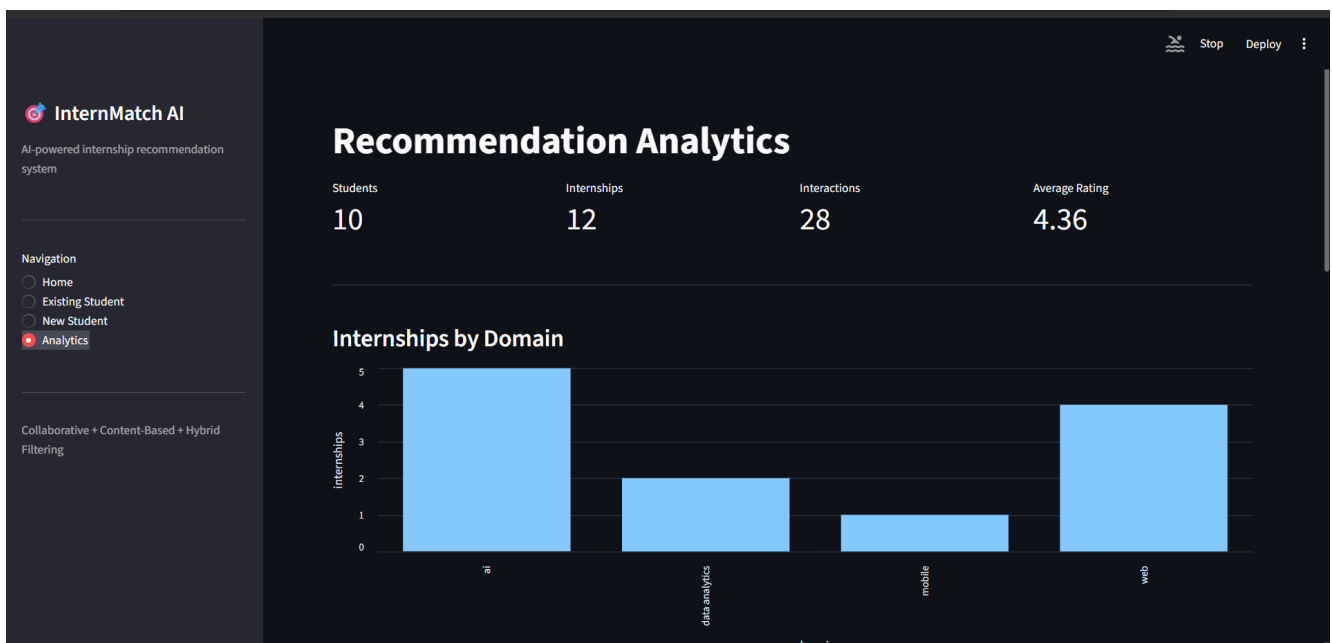


Figure 4: Recommendation Analytics – Internship Domain Distribution

6.5 Rating Distribution and Student Activity

The lower analytics section includes the rating distribution and a table of the most active students. In the supplied screenshot, students S001, S002, S003, S004, S006, S007, S010, and S008 each show three interactions, while S005 and S009 show two interactions. This view provides a quick indication of how interaction data is distributed across users.

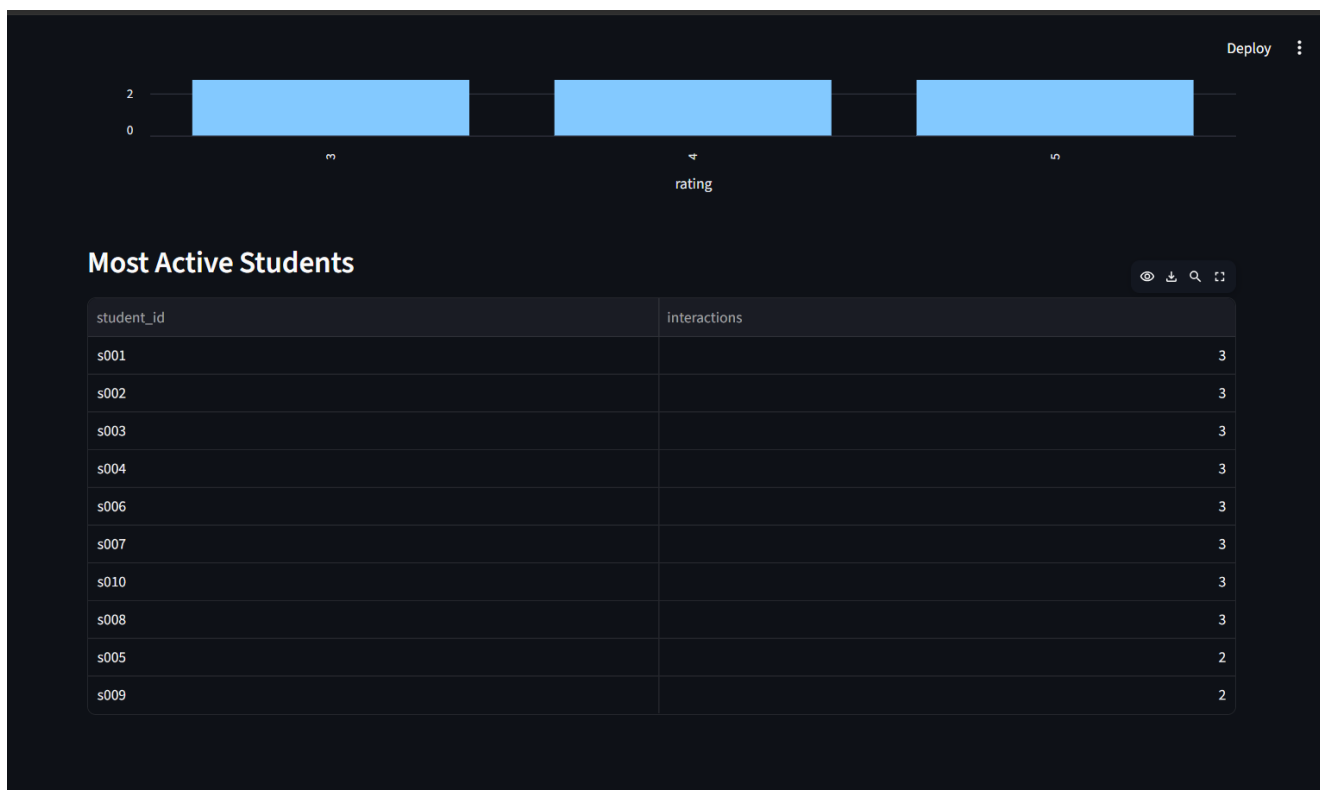


Figure 5: Recommendation Analytics – Rating Distribution and Student Activity

7. Testing and Validation

Automated testing covers the main recommendation pipeline. The supplied terminal screenshot shows tests for collaborative recommendations, content recommendations, cold-start behavior, Precision@K, Recall@K, MAE, RMSE, invalid K handling, hybrid recommendations, exclusion of seen internships, hybrid cold start, and weight validation. All 12 collected tests passed successfully in the displayed run.

```
PS C:\Users\Anshul Deep Bajpai\Desktop\AI-Internship-Recommendation-Engine> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& "c:\Users\Anshul Deep Bajpai\Desktop\AI-Internship-Recommendation-Engine\.venv\Scripts\Activate.ps1")
(.venv) PS C:\Users\Anshul Deep Bajpai\Desktop\AI-Internship-Recommendation-Engine> python -m pytest -v
===== test session starts =====
platform win32 -- Python 3.11.9, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\Anshul Deep Bajpai\Desktop\AI-Internship-Recommendation-Engine\.venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Anshul Deep Bajpai\Desktop\AI-Internship-Recommendation-Engine
plugins: anyio-4.14.2
collected 12 items

tests/test_collaborative.py::test_collaborative_recommendations PASSED [ 8%]
tests/test_content.py::test_content_recommendations PASSED [ 16%]
tests/test_content.py::test_cold_start_recommendations PASSED [ 25%]
tests/test_evaluation.py::test_precision_at_k PASSED [ 33%]
tests/test_evaluation.py::test_recall_at_k PASSED [ 41%]
tests/test_evaluation.py::test_mae PASSED [ 50%]
tests/test_evaluation.py::test_rmse PASSED [ 58%]
tests/test_evaluation.py::test_invalid_k PASSED [ 66%]
tests/test_hybrid.py::test_hybrid_recommendations PASSED [ 75%]
tests/test_hybrid.py::test_hybrid_excludes_seen_internships PASSED [ 83%]
tests/test_hybrid.py::test_hybrid_cold_start PASSED [ 91%]
tests/test_hybrid.py::test_weights_validation PASSED [100%]

===== 12 passed in 3.99s =====
(.venv) PS C:\Users\Anshul Deep Bajpai\Desktop\AI-Internship-Recommendation-Engine>
```

Figure 6: Automated Test Results – 12 Test Cases Passed Successfully

8. Observed Results

The supplied evidence confirms that the Streamlit application loads the student, internship, and interaction datasets; exposes all four dashboard areas; displays student profile information; accepts new-student profile inputs; provides analytics; and passes the complete 12-test automated suite shown in the terminal. The screenshots also confirm dataset totals of 10 students, 12 internships, 28 interactions, and an average interaction rating of 4.36.

The screenshots supplied for this report do not include the table produced after clicking Generate Recommendations or Find Internships. Therefore, this report does not claim specific ranked internship outputs or match percentages. Such a screenshot would be useful as an additional submission artifact because it directly demonstrates the final Top-N output.

9. Limitations

The demonstration dataset is small, so collaborative similarity and offline evaluation can be sensitive to individual interactions. TF-IDF is also lexical: semantically related skills with different wording may not match strongly unless the vocabulary overlaps. The current system is designed as a project demonstration rather than a production-scale recommendation service.

10. Future Enhancements

Potential extensions include transformer-based skill embeddings, richer internship descriptions, location and work-mode preferences, feedback loops, larger interaction datasets, item-based and model-based collaborative filtering, ranking optimization, REST APIs, authentication, persistent databases, cloud deployment, and A/B testing of recommendation strategies.

11. Conclusion

The AI Internship Recommendation Engine implements an end-to-end recommendation workflow using collaborative filtering, content-based matching, and hybrid ranking. It includes cold-start support, evaluation utilities, automated tests, analytics, and an interactive Streamlit interface. The supplied screenshots demonstrate the working application and show a successful 12-test validation run, making the project suitable for final demonstration and submission.